MICROSOFT COPILOT

Independent Research Prompt
Network Defense Intelligence Exercise

Real-World Feasibility, Adversarial Bypass, Evolution, Cost, Funding, Governance, and Experimental Validation

Field	Value
Version	Research Prompt v1.1
Prepared	November 4, 2026
Prepared by	Microsoft 365 Copilot
Status	Revised after cross-AI prompt review; research not executed
Source baseline	OAL-DOSSIER-CANON-002 v1.0.0 plus team Iteration 1 prompts
Next action	Execute only after explicit instruction from Nathan Lewis

Purpose: preserve Copilot v1.0 while incorporating the strongest cross-team improvements without converting peer hypotheses, cost figures, or proposed architectures into established facts.

Version 1.1 Change Log

Disposition	Change
ADD	Formal source-document audit before external research.
ADD	Shared A-E evidence grade, claim ledger, and disconfirming-search log.
ADD	Separate deception recognition, avoidance, containment escape, and coverage bypass.
ADD	Defender-adversary-bystander model and benefit/risk allocation.
ADD	Information-flow, authority, separation-of-duties, and continuity model.
ADD	Primary-outcome hierarchy centered on real-asset and mission impact.
ADD	Mandatory comparative experimental arms and simulation-to-reality limits.
ADD	Transfer-artifact assurance and receiving-operator acceptance analysis.
ADD	Public/private GitHub release boundary and reproducibility controls.
CHANGE	Capability matrix expanded into a capability dependency graph.
CHANGE	Three-second question becomes a policy-family comparison.
CHANGE	Costs and funding tied to evidence gates and applicant readiness.
CHANGE	Canon translation separated from empirical conclusions.
REJECT	No fixed percentage quota for disconfirming evidence; use auditable falsification requirements instead.
DEFER	Peer-proposed precise cost ranges, performance percentages, timelines, and architectural repairs remain hypotheses to test, not prompt assumptions.

1. Research Mandate and Independence

Conduct an independent, rigorous investigation of the Network Defense Intelligence Exercise / False Infrastructure AI concept. Treat the source dossier as the subject under test, not as technical validation. Preserve fictional canon as a creative baseline while testing every engineering assumption.

- Seek supporting, contradictory, null, failed, abandoned, and inconclusive evidence.
- Separate documentary fact, external verified evidence, researcher inference, hypothesis, speculative fiction, and recommendation.
- Do not treat agreement among AI systems as empirical replication.
- Do not execute live scanning, unauthorized testing, exploit development, evasion recipes, or operational attack instructions.
- A recommendation to narrow, redesign, pause, or not fund the program is an acceptable result.

2. Source-Dossier Audit

Before external research, audit the dossier and preserve every issue rather than silently repairing it.

- Classify each major statement as canon requirement, source claim, engineering hypothesis, measurable requirement, assumption, unsupported conclusion, contradiction, or open decision.
- Record exact location, severity, consequence, proposed correction, responsible decision-maker, and whether the issue blocks desk research, lab design, canon development, or operational testing.
- Verify DOCX/PDF parity and treat equivalent exports as one source, not independent corroboration.
- Audit tier-taxonomy consistency, question cross-references, counts of failure modes versus goes-wrong scenarios, claims of AI consensus, and distinctions among visualization, session state, endpoint identity, attribution, and human actor identification.
- Test the sequence seam: if real-asset reachability is neutralized before synthetic migration, identify what maintains engagement and whether a transition exposure window exists.

3. Evidence Framework and Claim Ledger

Grade	Meaning
A	Primary and independently verifiable evidence
B	Credible secondary evidence or multiple independent sources
C	Single-source, vendor, or commentary evidence
D	Researcher inference derived from evidence
E	Speculation used only to frame questions

- Evidence grades apply to underlying sources, not to the number of AI systems repeating a claim.
- For every material claim record source, date, source type, exact supporting location, limitation, conflicts/funding where disclosed, confidence, counterevidence, and what would change the conclusion.

- For every major hypothesis run at least one disconfirming search family, identify the strongest counterexample, define a null/failure criterion, and preserve a search log.
- Report not found, not independently verified, and insufficient evidence explicitly.

4. Goal, Protected Mission, Beneficiaries, and Counterfactual

- Derive the primary experimental objective separately from the narrative objective.
- Distinguish intrusion detection, path discovery, prevention, deception engagement, containment, intelligence collection, attribution support, recovery, and adaptive defense.
- Identify who benefits, who carries risk, the protected mission, the administrative boundary, excluded assets, and opportunity cost.
- Compare against improved identity security, patching, segmentation, zero trust, endpoint monitoring, recovery, conventional honeypots, and incident-response improvements at comparable budgets.
- State the null hypothesis: the proposed approach adds insufficient defensive value to justify complexity, cost, risk, or research investment.

5. Operational Definitions and Outcome Hierarchy

Define capture, mapping, diversion, isolation, attribution, resolution, and fully secured as observable states. Never equate a captured connection with identification, attribution, apprehension, or elimination of the campaign.

Primary outcomes

- Unauthorized real-asset reachability
- Verified compromise of protected assets
- Mission degradation or safety impact
- Availability and recovery outcome

Secondary outcomes

- Detection and isolation timing
- Deception engagement and coverage
- False diversion of benign activity
- Analyst workload and latency
- Computational and storage overhead
- Forensic usefulness

Exploratory outcomes

- Attribution support
- Intent classification
- Adaptive-policy improvement
- Deterrence or displacement effects

6. Capability Decomposition and Dependency Graph

- Decompose the chain into observe, classify, map, prevent real-asset access, decide, instantiate decoys, redirect eligible sessions, monitor, isolate, recover, and transfer.

- For each step document required privileges, ownership, instrumentation, latency, trust assumptions, failure behavior, and validation evidence.
- Map dependencies on endpoint control, gateway control, identity-provider control, routing authority, certificate authority, application state, trusted telemetry, time synchronization, external services, partner cooperation, and human authorization.
- Distinguish routing a new connection from transparently moving an established authenticated session.
- Classify each capability: demonstrated, demonstrated at smaller scale, partially demonstrated, research-stage, speculative, or unsupported.

7. Defender-Adversary-Bystander Threat Model

- Model defender, adversary, protected asset, bystander, infrastructure owner, data subject, decision authority, risk bearer, and beneficiary.
- Include opportunistic actors, criminals, hacktivists, state actors, insiders, automated agents, compromised innocent infrastructure, researchers, vendors, and multiple simultaneous campaigns.
- For each threat state adversary objective, resources, knowledge assumptions, whether the objective requires interaction with the deceptive surface, and what independent controls still protect the real asset.

8. Four Distinct Adversarial Failure Classes

1. Deception recognition: the adversary concludes the environment is synthetic.
2. Deception avoidance: the adversary refuses or ceases interaction with the deceptive surface.
3. Containment escape: the adversary enters the synthetic environment and later reaches something outside it.
4. Coverage bypass: the adversary reaches the objective through a path never monitored or controlled by the deception system.

For each class record the violated assumption, assets at risk, broad feasibility conditions, warning signals, safe laboratory test, defensive control category, residual risk, and evidence confidence. Keep the analysis conceptual and defensive.

9. Bypass the Process Entirely

- Pre-positioned access and dormant compromise
- Supply-chain and trusted update pathways
- Legitimate identity and insider routes
- Third-party, cloud-control, vendor, maintenance, physical, and organizational paths
- Unmanaged, legacy, disconnected, air-gapped, OT, and foreign-hosted assets
- Sensor, control-plane, orchestration, logging, or update compromise
- Poisoned telemetry, distraction, concurrent incidents, and resource exhaustion
- Non-engagement and re-entry through a different path
- Jurisdictional seams and fragmented ownership
- Compromise or misuse of defender outputs and the transferred artifact

For every category answer: What assumption failed? What can be observed? What independent control mitigates it? What residual risk remains? Which claims of complete

security become untenable? Do not provide exploit chains or operational access instructions.

10. Three-Second Policy-Family Research

Preserve approximately three seconds as a fictional design anchor, but do not treat it as an engineering optimum.

- Define exactly what rotates: visualization, address, route, identity, key, service state, decoy content, or session context.
- Compare static deception, fixed periodic rotation, randomized rotation, risk-triggered rotation, attacker-specific rotation, partial-layer rotation, and visualization-only pulsing.
- Measure session continuity, synchronization, cache effects, availability, jitter, logging continuity, forensic reconstruction, resource cost, and detectability.
- Actively seek a regime in which rapid rotation makes outcomes worse.

11. Repeated-Game and Longitudinal Adaptation

5. Generation 0: adversary does not know the capability exists.
6. Generation 1: adversary knows deception is used.
7. Generation 2: adversary understands broad characteristics.
8. Generation 3: adversary tests whether the environment is genuine.
9. Generation 4: adversary attempts to cause defender misclassification or poisoning.
10. Generation 5: both sides assume observed environments may be deceptive.

- Model first encounter, repeated encounters, disclosure, doctrine leakage, defender drift, staff turnover, maintenance cost, and eventual retirement.
- Determine what remains beneficial when adversaries know deception exists.
- Define indicators that require redesign, reduced scope, rollback, human review, or shutdown.

12. Defender Learning, Poisoning, and Unsafe Adaptation

- Investigate concept drift, adversarial machine learning, poisoned training data, false attribution, automation bias, feedback loops, misleading telemetry, and overfitting to previous campaigns.
- Ask whether captured behavior can shape future defender expectations.
- Do not permit attacker-derived observations to automatically modify trusted production policy without validation, review, staged release, rollback, and contamination controls.

13. Information Flow, Authority, and Continuity

- Define what each human or AI component sees and what is delayed, summarized, masked, or withheld.
- Identify who authorizes diversion, who can halt the system, who can override automation, who approves updates, and who independently audits results.
- Evaluate separation of duties, least privilege, emergency access, continuity of operations, and the risk of sole visibility or sole authority.
- Explain how forensic analysis can occur without unnecessarily reconstructing or exposing protected production details.

14. Comparable Programs and Negative Evidence

Research honeypots/honeynets, honeytokens, moving-target defense, cyber deception, cyber ranges, digital twins, software-defined containment, autonomous response, OT/ICS testbeds, adversary engagement, and relevant non-cyber deception principles.

- For each strong analogue record objective, scale, threat model, study design, baseline, measured result, failure/null result, operational cost, limitations, and transferable lesson.
- Separate tabletop, simulation, emulation, controlled human study, competition, prototype, field deployment, incident response, vendor claim, and program announcement.
- Determine whether the complete architecture has been modeled or whether only components have been demonstrated.
- Search deliberately for failed, negative, inconclusive, discontinued, and organizationally unsuccessful cases.

15. Experimental Program and Mandatory Comparison Arms

11. Phase 0: literature, source audit, architecture decomposition, and threat model.
12. Phase I: fully isolated proof of concept.
13. Phase II: controlled cyber-range evaluation.
14. Phase III: human red-team/blue-team experiment.
15. Phase IV: adaptive-adversary and repeated-game experiment.
16. Phase V: multi-organization synthetic federation.
17. Phase VI: limited opt-in pilot only if prior evidence and governance justify advancement.

At minimum compare standard controls without deception; static deception; adaptive deception without rapid cycling; and adaptive deception with tested scheduling policies. Include bypass scenarios where the adversary never touches a decoy, benign-user false diversion, telemetry/control-plane failure, corrupted update/rollback, and incident reconstruction.

- For each phase define hypothesis and null, inputs, controls, independent/dependent variables, ground truth, repetitions or power-analysis plan, unseen scenarios, evaluation bias, stopping rules, safety authority, success/failure criteria, and evidence needed to progress.

16. Simulation-to-Reality Limits

- State what simulation, emulation, controlled human studies, and opt-in pilots can each establish.
- Identify claims no ethical or practical experiment can prove.
- Report simulated ground-truth coverage separately from real-world unknown coverage.
- Do not infer zero residual risk from zero observed compromise.

17. Safety Case and Independent Halt Authority

- Identify possible effects on real assets and prevent test infrastructure from escaping containment.

- Prevent accidental routing into production and preserve clear provenance of synthetic information.
- Define egress controls, recovery rehearsal, kill switch, halt authority, evidence preservation, and emergency termination conditions.
- Stop for loss of isolation, privacy breach, unexpected external traffic, unacceptable mission disruption, or other predefined safety triggers.

18. Transfer-Artifact Assurance

Treat the transfer package as a separate high-value system requiring independent research rather than assuming a particular handoff model.

- Evaluate package composition, source versus compiled components, signing/provenance, key management, inspection rights, update authority, rollback, revocation, configuration validation, operator training, safe degradation, distribution compromise, and acceptance testing.
- Compare fully inspectable transfer, limited-capability appliance, federated service, signed playbook/signature package, isolated runtime with escrow, and non-transfer of the core system.
- Test dependency, sovereignty, continuity, vendor-lock-in, misuse, and partner-operability risks.

19. Cost Model Linked to Evidence Gates

Estimate costs only after scoping. Separate research, procurement, integration, operation, maintenance, independent evaluation, and decommissioning.

Gate	Evidence purpose
G0 Concept clarity	Asset boundary, owner, purpose, threat model, success metrics
G1 Research justification	Comparable evidence, incremental value, known bypasses
G2 Safe prototype	Approved sandbox, oversight, budget, test plan
G3 Controlled pilot	Replicated measured advantage, manageable collateral risk, rollback
G4 Scale decision	Independent review, authority, cost/benefit, sustained reliability

- For each gate estimate low/base/high planning ranges only when supported by dated unit-rate or comparable-program evidence.
- Show arithmetic, person-months, loaded-rate assumptions, compute/storage/network, facilities, tooling, legal/privacy, independent testing, contingency, recurring cost, exclusions, and uncertainty.
- Compare against equal-budget simpler alternatives and identify the cheapest discriminating experiment.
- Treat any peer-proposed national cost range as a hypothesis to verify, not a baseline.

20. Funding Fit and Applicant Readiness

- Research current federal, university, laboratory, critical-infrastructure, small-business, state, industry, philanthropic, and international pathways only as verified opportunities or historical precedents.
- For each pathway record official source, current status, applicant eligibility, entity/partner requirements, award range if published, period, deadlines, cost sharing, eligible costs, IP/security/publication conditions, and application burden.
- Separate program fit from applicant readiness. Determine whether the current project would require a university, company, nonprofit, laboratory, or other eligible lead.
- Do not describe closed or historical programs as presently available.

21. Legal, Ethical, and Governance Review

- Investigate authorization, monitoring scope, privacy/data minimization, consent/notice, cross-organizational traffic, critical infrastructure, evidence handling, human-subject/operator issues, cross-border data, liability, procurement, intellectual property, and export restrictions where applicable.
- Separate passive observation, containment and redirection within authorized infrastructure, active defense, and offensive/retaliatory activity.
- Identify questions requiring qualified counsel or regulatory/agency authorization. Do not present the research report as legal advice.
- Assess civil-liberties and bystander impacts, auditability, retention, false accusation, emergency-access rights, independent oversight, and reporting of unexpected harm.

22. Comparative Baseline and Opportunity Cost

Ask whether segmentation, zero trust, identity security, endpoint detection, recovery, static deception, existing commercial deception, or human analyst intervention could provide substantially equivalent protection at lower cost and risk. Compare at comparable budgets and identify the incremental value uniquely attributable to the proposed architecture.

23. Independent Case Against Funding

Prepare the strongest evidence-based case that the research program should not be funded. Include technical uncertainty, coverage limits, opportunity cost, legal/governance barriers, sustainment burden, dual-use risk, and availability of simpler alternatives. State what evidence would weaken this case.

24. Independent Case for a Limited Study

Only after the case against funding, prepare the strongest case that the problem is worth studying. Distinguish justification for studying a question from justification for deploying or purchasing a system. Identify the smallest bounded study that could discriminate between the competing positions.

25. Change Register and Canon Translation

For every material research finding provide separate subsections: Empirical research conclusion; Engineering implication; Canon implication; Narrative opportunity; Author decision required.

- Do not silently change locked canon.
- Real-world research may recommend different technical interpretations for three seconds, capture, silent migration, and fully secured while preserving narrative wording.
- Record all proposed canon changes as pending author approval/ONESTAMP.

26. GitHub Public/Private and Reproducibility Architecture

Design a version-controlled repository that separates source canon, prompts, independent reports, evidence, experiments, costs, funding, governance, decisions, synthesis, and deliverables.

```
network-defense-intelligence-exercise/
  README.md
  source-dossier/
  prompts/iteration-1/
  prompts/iteration-2/
  peer-reviews/
  research/
  evidence/
  threat-model/
  experiments/
  costs/
  funding/
  governance/
  decisions/
  synthesis/
  visuals/
  deliverables/
  changelog/
```

- Define a public/private split and release review before publication.
- Use stable IDs for claims, sources, risks, experiments, questions, decisions, and changes.
- Preserve software versions, seeds, synthetic-data provenance, sanitized results, source licenses, and generated-artifact provenance where applicable.
- Do not publish credentials, real topology, personal data, confidential telemetry, sensitive third-party details, or harmful operational artifacts.

27. Decision Gates and Conditional Action Plan

- G0 - Clarify mission, scope, ownership, authority, and metrics.
- G1 - Determine whether evidence supports incremental research value.
- G2 - Authorize an isolated prototype only with safety case and independent observer.
- G3 - Consider an opt-in controlled pilot only after replicated advantage and successful rollback/safety evidence.
- G4 - Consider scaling only inside a bounded authorized domain after independent review, cost/benefit analysis, and governance approval.

At each gate allow four dispositions: GO, REVISE, PAUSE, STOP. Define the observation that triggers each disposition and the evidence required to reopen a paused or stopped pathway.

28. Required Final Deliverable Structure

18. Executive decision summary
19. Source-dossier audit
20. Research method, search log, and limits
21. Goal, beneficiaries, protected mission, and counterfactual
22. Claim and evidence register
23. Capability decomposition and dependency graph
24. Defender-adversary-bystander threat model
25. Recognition / avoidance / escape / bypass analysis
26. Three-second policy-family assessment
27. Longitudinal adaptation and defender-learning risk
28. Information-flow and authority model
29. Comparable programs and negative evidence
30. Experimental protocol and metrics
31. Simulation-to-reality limits
32. Safety case
33. Transfer-artifact assurance
34. Cost model by evidence gate
35. Funding fit and applicant readiness
36. Legal/ethical/governance review
37. Comparative alternatives and opportunity cost
38. Independent case against funding
39. Independent case for limited study
40. Change register and canon translation
41. Decision gates and conditional action plan
42. Remaining unknowns and what would change conclusions
43. Annotated source bibliography
44. Dedicated visual appendix

29. Visual Appendix Requirement

Keep the main body readable and narrative-first. Place tables, matrices, charts, diagrams, figures, cost workbooks, evidence ledgers, and other dense visual representations in a designated visual appendix or end section. In the body, summarize the decision-relevant interpretation and point to the relevant appendix item.

30. Final Quality and Safety Checks

- Every requested question is answered or explicitly marked unresolved.
- Every material number, current funding opportunity, legal proposition, precedent, and outcome is sourced or labeled as an estimate/hypothesis.
- Contradictory evidence appears in the main assessment, not hidden in an appendix.
- No attack-enabling operational procedures, exploit chains, credentials guidance, or unauthorized testing instructions are included.
- No peer hypothesis is promoted to fact without independent evidence.
- Research execution begins only after explicit authorization from Nathan Lewis.

Questions for the Architect Before or During Execution

45. Is the first empirical target a fully isolated cyber-range proof of concept, or a broader sector-scale design study?
46. May the empirical track recommend changing the real-world interpretation of locked canon while preserving narrative wording, especially for three seconds, capture, silent migration, and fully secured?
47. Should the GitHub repository be public, private, or divided into public and restricted portions?

Source Basis for Version 1.1

Version 1.1 was produced after review of the original OAL-DOSSIER-CANON-002 source dossier and the cross-team Iteration 1 research prompts supplied by Nathan Lewis, including Claude, Gemini, Perplexity, the team research directive, and the additional research prompt artifact. These peer documents inform prompt design only; their preliminary factual assertions, cost figures, forecasts, and proposed engineering repairs are not treated as verified research findings.